included NO_2 and O_3, respectively, to examine the robustness of the main effects. Co-pollutants were matched to the $PM_{2.5}$ exposures in metric, lag, and linearity; meaning daily maximum $PM_{2.5}$ was fit with daily maximum NO_2 or O_3. $PM_{2.5}$ DECH-9, -12, and − 15 were fit with NO_2 DECH-5 and DECH-13 (ppb•hour) or O_3 DECH-30 (ppb•hour). Thresholds were selected based on the 2021 WHO Global Air Quality Guideline for these pollutants (WHO 2021), which is 10 and 25 µg/m^3 (5 and 13 ppb) for NO_2 and 60 (30 ppb) µg/m^3 for O_3. Co-pollutants assumed the same lag period as $PM_{2.5}$ and a linear relationship.

Table 1 Summary statistics of cluster-level demographics, outcomes, and exposures

Demographics	By cluster	
	Mean (SD)[2]	Median (Min, Max)
# of ZCTAs (*N* = 455)	13 (11.8)	10 (2, 55)
Population Count	477,655 (397,638)	383,934 (40,258, 1,531,522)
% Black	6.5% (8.1%)	4.9% (0.5%, 59.6%)
% Hispanic	46.8% (22.1%)	49.1% (15.6%, 92.3%)
% population with household income less than 2x FPL	32.9% (12.6%)	36.5% (17.0%, 58.6%)
Median age (years)	36.8 (4.2)	36.2 (29.7, 53.6)
Health Outcome(s) (per 100k population)	By Cluster-day	
	Mean (SD)	Median (Min, Max)
# of Cardiovascular-related EDVs	4.9 (2.2)	4.5 (0.4, 25.8)
# of Respiratory-related EDVs	10.7 (6.4)	9.0 (0.5, 54.1)
Exposure(s)	Mean (SD)	IQR
$PM_{2.5}$		
DECH-9 (µg/m^3•hour)	89.3 (106.7)	110
DECH-12 (µg/m^3•hour)	56.7 (88.7)	65.8
DECH-15 (µg/m^3•hour)	35.8 (72.2)	32.7
Daily (24-hour) maximum (µg/m^3)	19.3 (13.9)	10.4
Daily (24-hour) average (µg/m^3)	11.2 (5.6)	6.8
Apparent Temperature (°F)	63.6 (10.7)	16.4
O_3		
DECH-30 (ppb•hour)	171.8 (137.6)	180.1
Daytime 8-hour maximum[1] (ppb)	46.0 (13.8)	16.1
Daily (24-hour) average (ppb)	29.9 (9.8)	13.7
NO_2		
DECH-5 (ppb•hour)	179.8 (154.1)	212.7
DECH-13 (ppb•hour)	79.9 (104.1)	111.9
Daily (24-hour) maximum (ppb)	21.9 (10.1)	14.9
Daily (24-hour) average (ppb)	12.3 (6.9)	9.7

[1] Maximum of 8-hour average O_3.

[2] Population weighted mean and standard deviation were taken except # of ZCTAs, population count, and health outcomes, which is estimated by mean and standard deviation.

Results

Summary statistics

The final study population consisted of 35 regional clusters, with a 35,000 minimum population constraint for each cluster, constructed from the 455 ZCTAs in the South Coast AQMD domain (Supplemental Fig. 1). Each cluster covered 2 to 55 ZCTAs and has an average population of 477,655. Table 1 summarized the sociodemographic characteristics of the population in the study area at the cluster level. On average, 6.5% of residents reported non-Hispanic Black race, 46.8% reported Hispanic ethnicity, and 32.9% of households reported incomes of less than two times the federal poverty level. The population-weighted cluster median age was 36.8 years old.

Each cluster had 699 unique days of exposure and outcome data, from May of 2018 to March of 2020. The population weighted means (standard deviation) of DECH-9, DECH-12, DECH-15, daily maximum, and daily average $PM_{2.5}$ level were 89.3 (106.7) µg/m^3•hour, 56.7 (88.7) µg/m^3•hour, 35.8 (72.2) µg/m^3•hour, 19.3 (13.9) µg/m^3, and 11.2 (5.6) µg/m^3, respectively. Subdaily and daily metrics of O_3 and NO_2 levels were also calculated. Zero DECH values were observed on many days, specifically, 6.5%, 15.8%, and 26.5% of observations across all clusters for $PM_{2.5}$ DECH-9, -12, and − 15, 2.5% for O_3 DECH-30, and 1.4% and 17.3% for NO_2 DECH-5 and − 13, respectively. Concentrations of $PM_{2.5}$, NO_2, O_3, and apparent temperature were weakly correlated (Spearman correlation coefficients < 0.4).

The average daily counts of cardiovascular- and respiratory-related EDVs were 4.9 (2.2) and 10.7 (6.4) per 100,000 population per cluster, respectively. When examining trends over time, cardiovascular-related EDVs showed little variation in distribution over the study period, whereas respiratory EDVs peaked during colder months.

Associations between daily EDVs and subdaily exposure metrics

The associations between daily EDVs and $PM_{2.5}$ metrics were quantified as excess risk per interquartile range (IQR) increase in each exposure metric (Table 1). Figure 1 showed risk estimates of cardiovascular- and respiratory-related EDVs for each subdaily $PM_{2.5}$ metric at different lags.

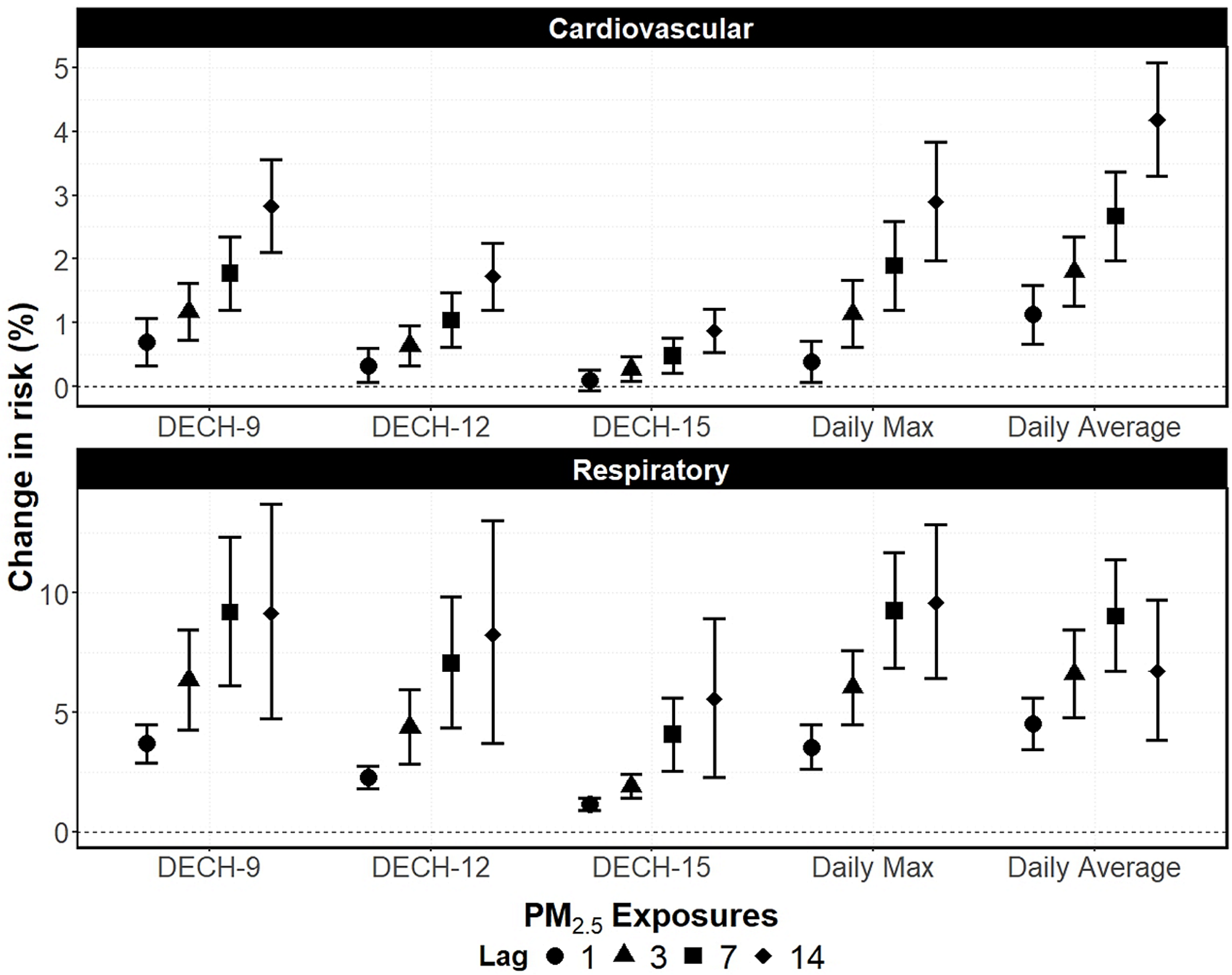

Fig. 1 Excess risk of cardiovascular- and respiratory-related EDVs and confidence intervals associated with an IQR increase in $PM_{2.5}$ exposure metrics at different distributed lags (exposure during the prior 1, 3, 7, 14 days to an EDV), Southern California (2018–2020)

The daily average concentration metric for $PM_{2.5}$ was also included for reference.

All subdaily $PM_{2.5}$ metrics were associated with elevated risk of cardiovascular-related EDVs, consistent with the daily $PM_{2.5}$ metric results (Fig. 1). For example, risk for cardiovascular-related EDVs increased by 1.77% (95% CI: 1.20, 2.34), 1.04% (0.61, 1.47), and 1.89% (1.21, 2.59) per IQR increase of DECH-9, DECH-12, and daily maximum $PM_{2.5}$ during 7-day lag period, respectively. Of the subdaily metrics, daily maximum $PM_{2.5}$ and DECH-9 showed greatest excess risks, most closely mirroring the associations for daily average $PM_{2.5}$. The associations for DECH-12 and DECH-15 were noticeably attenuated compared to DECH-9, possibly due to a large number of zero or near-zero values, as mentioned earlier.

Similarly, all subdaily $PM_{2.5}$ metrics were positively associated with respiratory EDVs across all lags. For example, risk of respiratory EDVs increased by 6.34% (4.25, 8.48), 4.39% (2.85, 5.95), 1.91% (1.41, 2.41), 6.04% (4.51, 7.60), and 6.61% (4.78, 8.47) per each IQR increase of DECH-9, DECH-12, DECH-15, daily maximum $PM_{2.5}$ and daily average $PM_{2.5}$ during 3-day lag period. Furthermore, in contrast with the daily exposure metric, where excess respiratory EDV risk peaked with distributed 7-day lag but attenuated with 14-day lag, risk estimates of subdaily metrics remained or slightly increased with the longer lag than 7-day. This may suggest that the elevated risks from subdaily exposure on the respiratory system last longer than that of the daily metric.

Out of distributed lags tested, models with 7-day lags and 3-day lags had minimal qAIC for cardiovascular- and respiratory-related outcomes, respectively. Therefore, we present results of lag 7 for cardiovascular-related EDVs and lag 3 for respiratory EDVs for the following stratified and sensitivity analysis.

Stratified analyses

Stratification by age group

When stratifying by patient age, older adults (ages 65 + years) showed higher excess risks for cardiovascular-related outcomes across different $PM_{2.5}$ metrics compared to adults (ages 18–64 years), though differences were not statistically significant (Fig. 2).

For respiratory outcomes, children exhibited higher excess risks than both adult groups across all metrics. Differences were statistically significant for DECH metrics in tests for heterogeneity but not for the daily average $PM_{2.5}$ and daily maximum $PM_{2.5}$, though the effect sizes of DECH metrics were smaller than those of daily average and maximum $PM_{2.5}$. Older adults also exhibited higher risks than adults; the differences were significant for $PM_{2.5}$ DECH-9, DECH-12, and daily average.

Again, for both types of outcomes, daily maximum and DECH-9 estimates resembled the association for daily average metric more closely. DECH-12 and DECH-15 showed smaller excess risk estimates, partially due to large percentage of zeros among DECH-12 and − 15 values.

Stratification by race/ethnicity

We observed a slightly higher excess risk of cardiovascular-related EDVs among non-Hispanic White populations and a slightly higher excess risk of respiratory EDVs among Hispanic populations, although neither modification was

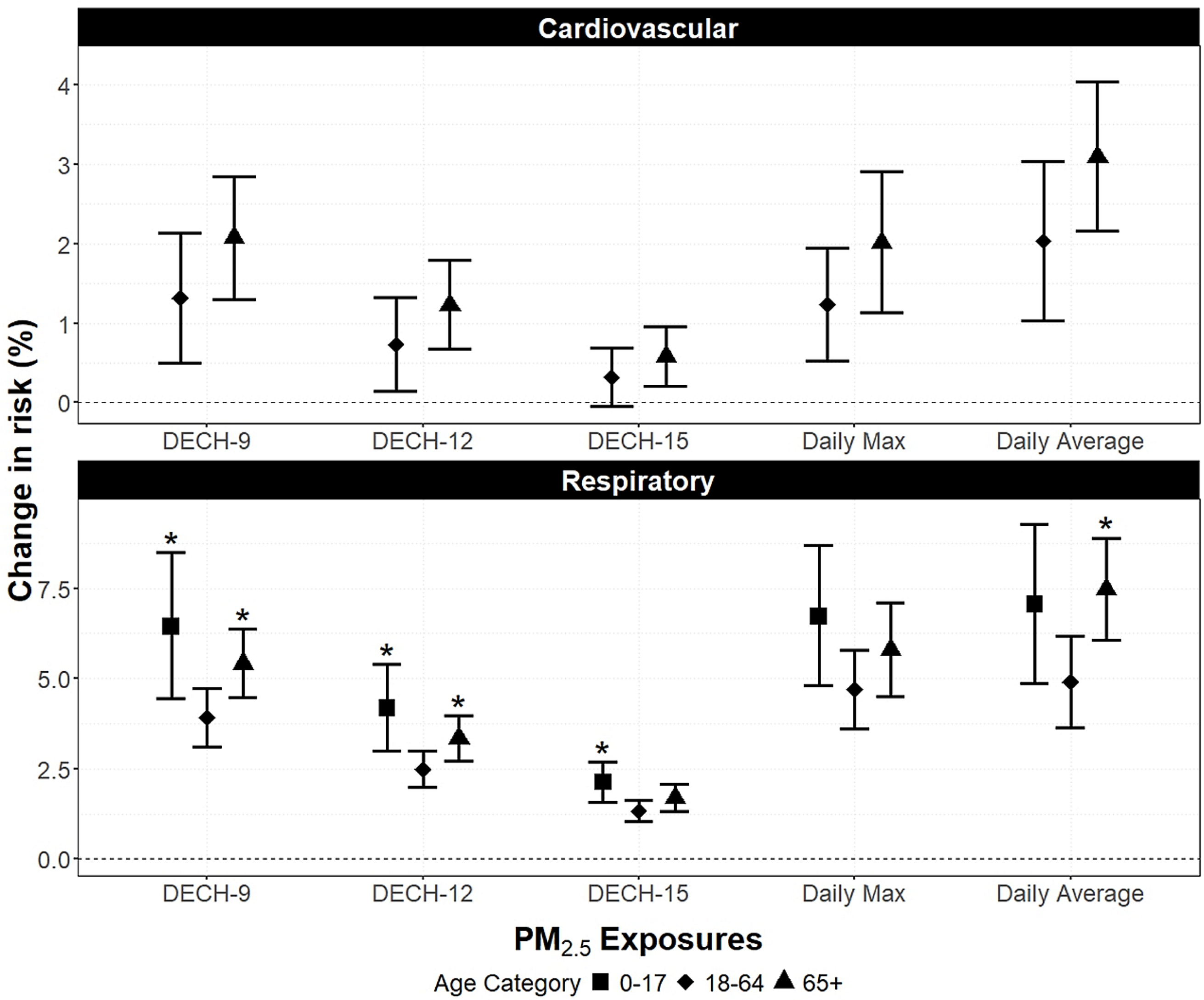

Fig. 2 Excess risk and confidence intervals of cardiovascular- and respiratory-related EDVs per each IQR increase of $PM_{2.5}$ exposure metrics, stratified by patient age group. Asterisk (*) represents statistically significant difference compared to the reference group, age 18–64. For cardiovascular-related outcomes, results of a 7-day lag were presented; the children category was not included given the low outcome counts. For respiratory outcomes, results of a 3-day lag were presented

statistically significant (Fig. 3). The variation in the risk estimates of respiratory EDVs between the non-Hispanic White and Hispanic group may be partly explained by the average ages of patients between strata (43.3 years for non-Hispanic White vs. 23.4 years for Hispanic). In contrast, the average ages of cardiovascular-related patients were similar between the two race/ethnicity groups. The direction and patterns of effect measure modification were consistent across subdaily metrics, as well as daily average $PM_{2.5}$.

Stratification by cluster poverty level

Given the strong modification effect by age group, the poverty stratification was conducted at the cluster level in addition to the age group stratification for each cluster (double stratification). We divided clusters into three poverty-based strata: low (N of clusters = 9, on average 19.3% of population with household income less than two times of federal poverty line); moderate (N = 12, 35.1%); high (N = 14, 49.5%) (Supplemental Table 1). Low poverty clusters had the lowest percentages of residents reporting Black race (3.8 ± 1.6%) and Hispanic ethnicity (25.1 ± 8.8%) and the highest median age (40.1 ± 1.6 years old) among the three strata. High poverty clusters had the highest percentage of Black residents (9.1 ± 8.2%), Hispanic residents (68.5 ± 14.9%), and youngest median age (32.9 ± 2.9 years old).

For cardiovascular-related EDVs, both adult groups living in low poverty clusters had higher excess risks than the moderate and high poverty groups across all $PM_{2.5}$ metrics (Fig. 4). For respiratory EDVs, older adults

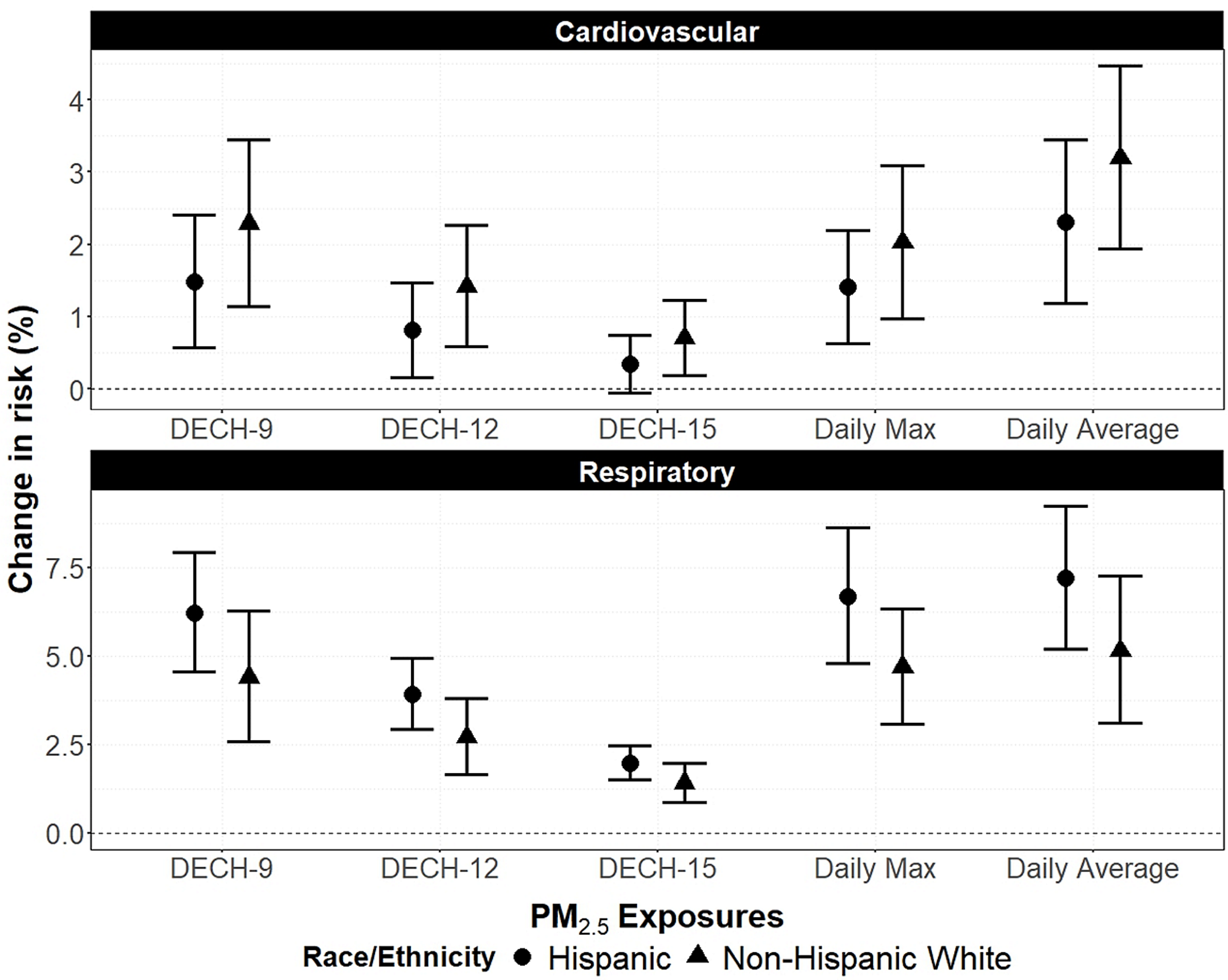

Fig. 3 Excess risk and confidence intervals for the association between $PM_{2.5}$ exposure metrics and EDVs per each IQR increase of $PM_{2.5}$ exposure metrics, stratified by patient race/ethnicity and outcome type. For cardiovascular-related outcomes, results of a 7-day lag were presented; for respiratory outcomes, results of a 3-day lag were presented. None of the effect modification shown in this figure was statistically significant.